Project Proposal

Title

VR-Based Geometric Transformation Training Environment for 3D Spatial Reasoning

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Introduction / Problem Statement

Spatial transformations such as translation, rotation, and scaling form the mathematical backbone of robotics, computer graphics, AR/VR systems, and surgical navigation. Despite their centrality, these concepts are still taught and trained primarily using two-dimensional diagrams, symbolic matrices, and static textbook illustrations. This creates a a gap between how transformations are represented and how they are applied in three-dimensional space.

In professional domains where precise spatial reasoning is critical, this mismatch leads to persistent errors in coordinate frame alignment, calibration, and spatial planning. These errors are not caused by lack of mathematical knowledge, but by insufficient 3D intuition developed through 2D training interfaces. Current training standards rely on symbolic manipulation rather than embodied spatial interaction, which often makes it difficult to build correct geometric intuition.

This project proposes a virtual reality training environment that replaces 2D-based instruction with a native 3D interaction paradigm. By allowing users to directly apply and observe geometric transformations in immersive space, the system addresses limitations of current training methods and proposes an alternative approach for learning spatial transformations and early professional training.

Objectives

Primary Objective - To design and implement a VR environment that enables direct manipulation and visualization of geometric transformations in 3D space.

Specific Objectives - To implement translation, rotation, and uniform scaling on a single 3D object using VR controllers. - To visualize local and global coordinate frames during each transformation. - To display the corresponding transformation matrix in real time for each operation. - To observe whether users can better predict transformation outcomes after interaction.

Methodology

System Architecture & Tech Stack

- Development Engine: Unity 3D
- Programming Language: C#
- VR Framework: OpenXR
- Target Hardware: Standalone VR headset or PC-tethered headset

The system consists of three main modules: Interaction Module, Transformation Engine, and Visualization Module.

Interaction Module

User input is obtained through VR controllers. Each controller input is mapped to a single transformation mode: - Translation mode: joystick-based movement along X, Y, Z axes - Rotation mode: controller orientation mapped to object rotation - Scaling mode: trigger-based uniform scaling

Only one transformation mode is active at a time to avoid compound complexity.

Transformation Engine

For each interaction, a 4x4 homogeneous transformation matrix is constructed and applied to the object's local coordinate frame. Transformations are implemented using explicit matrix multiplication rather than engine shortcuts to keep the mathematical process clear.

The system maintains: - Local object coordinates - World coordinates - Current transformation matrix

Visualization Module

- Render local and global coordinate axes as colored lines
- Display the current transformation matrix as a floating UI panel
- Show pre- and post-transformation object poses side-by-side for comparison

No physics simulation, lighting estimation, or deformation is included to keep the system simple and predictable.

Testing & Validation

- Unit testing for each transformation operation
- Visual verification of matrix correctness
- User task: predict final pose before applying transformation

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Timeline (3 Weeks)

Week	Tasks	Deliverables

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1	Unity setup, VR integration, base scene setup	Working VR scene with object interaction
2	Implement transformations and coordinate visualization	Translation, rotation, scaling functional
3	UI overlay, testing, documentation	Final prototype and demo-ready build

Resources

Human Resources - Team of 2–3 students - Roles: Interaction development, math/logic implementation, testing and documentation

Software Resources - Unity 3D (free academic license) - OpenXR SDK

Hardware Resources - One VR headset - Development PC

No external funding or paid software required.

Feasibility & Risk Control

- Scope restricted to a single object and three transformations
- No physics, animation, or ML components
- Offline, single-user system only
- Backup plan: desktop 3D version if VR hardware access is limited